Lecture 11. Causal Inference

Potential outcomes, randomization, confounding, matching and weighting

Contents

11.1 Why causation **2**

11.2 Potential outcomes **2**

11.3 Randomised experiments and observational studies **3**

11.3.1 Unconfoundedness 4

11.4 Propensity score methods **4**

11.4.1 Stratification 5

11.4.2 Matching 5

11.4.3 Inverse probability weighting 5

11.5 Example: job training and earnings **6**

11.1 Why causation

Motivation. Prediction asks what Y will be; many real questions ask what Y would be if we changed something. Answering that from data requires more than a good model: it requires assumptions about how treatment was assigned, and designs that make those assumptions believable.

Some questions are descriptive, but many, especially in medicine, are causal: Is smoking harmful? Would lower air pollution reduce deaths from respiratory disease? Does delivery at high-level neonatal intensive care units lower the risk of lung disease (BPD) as well as death? What proportion of fever cases is attributable to malaria?

A **confounder** X affects both the treatment Z and the outcome Y. Ice-cream sales and drownings rise together because hot weather drives both; often, however, we do not recognise all confounders.

Job training (LaLonde). A job-training programme in 1976: did it raise earnings in 1978? Treated and control people differ greatly before treatment:

```
lalonde %>%
  group_by(treat) %>%
  summarise(across(c(age, educ, married, re74, re75, re78), mean), black = mean(race ==
    "black"))
```

```
## # A tibble: 2 x 8
##   treat   age  educ married  re74  re75  re78 black
##   <int> <dbl> <dbl>   <dbl> <dbl> <dbl> <dbl> <dbl>
## 1     0  28.0  10.2   0.513 5619. 2466. 6984. 0.203
## 2     1  25.8  10.3   0.189 2096. 1532. 6349. 0.843
```

The treated are much more often Black, less often married, and earned far less in 1974–75. Is any difference in 1978 earnings due to the programme or to these factors?

Smoking and lung cancer. By the mid-1940s lung cancer had tripled in three decades, with candidate explanations including air pollution from cars and tar roads, an ageing population, radiography, and better diagnosis. In 1950 observational studies appeared (Wynder and Graham; Doll and Hill; Hammond). Hammond matched 36 975 heavy smokers to non-smokers on age, race, residence, occupational exposure, religion, education, marital status, alcohol, sleep, exercise and more. In 122 pairs exactly one person died of lung cancer: in 110 the smoker, in 12 the non-smoker. With no effect, each discordant pair is equally likely to go either way, so the number X of smoker deaths is $B(122, 1/2)$:

```
binom.test(110, 122, p = 0.5, alternative = "greater")$p.value
```

```
## [1] 2.733658e-21
```

The inference is simple; what could be wrong is an *unmeasured* confounder. R. A. Fisher, a heavy smoker, argued that a gene might cause both smoking and cancer. Association is not causation, but causal reasoning is hard.

How people reason about causes. Zeus receives a heart transplant and dies five days later. If we could know that without the transplant he would have been alive after five days, we would say the transplant caused his death. Causal effects compare what happened with what *would have happened*.

11.2 Potential outcomes

Motivation. The comparison "with versus without treatment" needs notation in which both possibilities exist. Potential outcomes provide it, and they make precise both what we want to estimate and why it cannot be observed directly.

For units $i = 1, \ldots, N$: covariates X_i; treatment $Z_i \in \{0, 1\}$; **potential outcomes** $Y_i(1)$ (if treated) and $Y_i(0)$ (if not); and the observed outcome

$$Y_i = Y_i(Z_i) = Z_i Y_i(1) + (1 - Z_i) Y_i(0).$$

For Zeus, $Z_i = 1$ and $Y_i = 1$ (death), with $Y_i(1) = 1$, $Y_i(0) = 0$: $Y_i(1)$ is the *factual* outcome and $Y_i(0)$ the *counterfactual* one, which can only be imagined. (Other texts write treatment A and potential outcomes $Y^{a=1}$, $Y^{a=0}$.)

The notation hides assumptions:

- **no interference**: Y_i depends only on unit i's own treatment, not on Z_j or Y_j for $j \neq i$;
- **consistency**: $Y_i = Y_i(1)$ if $Z_i = 1$, which requires **no multiple versions of treatment**: all treated units receive the same treatment (e.g. placebo patients must receive nothing else).

Together these are the **stable unit treatment value assumption** (SUTVA).

Causal effects. The individual effect is $\tau_i = Y_i(1) - Y_i(0)$; the **average treatment effect** is

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)] = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)].$$

Other measures: the risk ratio $\mathbb{E}[Y_i(1)]/\mathbb{E}[Y_i(0)]$ and, for binary outcomes, the odds ratio $\dfrac{P(Y_i(1) = 1)\, P(Y_i(0) = 0}{P(Y_i(1) = 0)\, P(Y_i(0) = 1}$

The **sample** ATE is $\frac{1}{N} \sum_i \{Y_i(1) - Y_i(0)\}$, the **population** ATE is $\mathbb{E}[Y_i(1) - Y_i(0)]$, and SATE $\to$ PATE as $N \to \infty$.

Science table versus statistician's table. The "science table" lists both potential outcomes and the covariates for every unit; the data reveal only $Y_i(1)$ for treated units and $Y_i(0)$ for controls, the other entry being "?" (covariates are fully observed unless missing).

The fundamental problem of causal inference (Holland, 1986): τ_i is never observed, since only one potential outcome is. Individual effects cannot be estimated, but under assumptions on the treatment assignment the ATE $\mathbb{E}[\tau_i] = \mathbb{E}[Y_i(1)] - \mathbb{E}[Y_i(0)]$ can: random assignment (randomised experiments) or unconfoundedness (observational studies), together with **overlap**, $0 < P(Z = 1 \mid X) < 1$: every unit has a chance of receiving and of not receiving treatment.

11.3 Randomised experiments and observational studies

Motivation. Randomisation makes the treated and control groups alike in every respect, measured or not, so their difference in means is the causal effect. Without randomisation, we must create comparable groups ourselves, using the covariates.

Randomised experiments (clinical trials, A/B tests) are the gold standard but often infeasible; observational studies (census, shopping histories, streaming data) are easy to obtain but hard to analyse.

Under random assignment, treated and control groups tend to be similar in both observed and unobserved covariates; randomisation gives unbiased estimators and is the "reasoned basis for inference" (Fisher, 1935), since the physical act of randomising induces the distribution used for exact tests. Formally, random assignment gives

$$(Y(1), Y(0)) \perp\!\!\!\perp Z.$$

In a cholesterol-drug trial with assignment by a fair coin, people's potential cholesterol levels have nothing to do with the coin, so $\mathbb{E}[Y(1)] = \mathbb{E}[Y(1) \mid Z = 1] = \mathbb{E}[Y(1) \mid Z = 0]$, and $Y(1)$ is observed when $Z = 1$. Hence

$$\mathbb{E}[Y(1) - Y(0)] = \mathbb{E}[Y(1) \mid Z = 1] - \mathbb{E}[Y(0) \mid Z = 0] = \mathbb{E}[Y \mid Z = 1] - \mathbb{E}[Y \mid Z = 0] = \bar{Y}_t - \bar{Y}_c,$$

the difference in group means, with no assumption on the distributions of $Y(1), Y(0)$. Independence of potential outcomes and treatment is the **identification assumption**.

When it fails. If a doctor gives the drug to patients expected to have high cholesterol without it, higher $Y_i(0)$ means a higher chance of $Z_i = 1$, so $\mathbb{E}[Y(0) \mid Z = 1] > \mathbb{E}[Y(0) \mid Z = 0]$ and the naive comparison is biased. The doctor's decision is based on medical history X, a confounder.

In an observational study treated and control groups differ systematically. **Overt bias** from observed covariates is removed by adjustment (regression, matching, weighting); **hidden bias** from unobserved covariates needs special designs (instrumental variables, discontinuities, difference-in-differences) and sensitivity analysis.

11.3.1 Unconfoundedness

The key assumption is

$$(Y(1), Y(0)) \perp\!\!\!\perp Z \mid X \qquad \text{(unconfoundedness, ignorability)}:$$

given enough information X, treatment looks random among units with the same X. Then

$$\mathbb{E}[Y(1)] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid X]\big] = \mathbb{E}\big[\mathbb{E}[Y(1) \mid X, Z = 1]\big] = \mathbb{E}\big[\mathbb{E}[Y \mid X, Z = 1]\big],$$

and likewise for $Y(0)$ with $Z = 0$. So fit an outcome model on the treated, $\hat{\mu}_1(X)$, another on the controls, $\hat{\mu}_0(X)$, and estimate

$$\hat{\tau}^{\text{ATE}} = \frac{1}{n} \sum_{i=1}^{n} \{\hat{\mu}_1(X_i) - \hat{\mu}_0(X_i)\}.$$

In general $\mathbb{E}[Y(1)] \neq \mathbb{E}[Y \mid Z = 1]$: causation and association differ.

Connection to regression. The classical adjustment $\mathbb{E}(Y \mid X) = \beta_0 + \beta_1 Z + \beta_2 X$ corresponds to $\mathbb{E}[Y(1) \mid X] = \beta_0 + \beta_1 + \beta_2 X$ and $\mathbb{E}[Y(0) \mid X] = \beta_0 + \beta_2 X$, so the ATE is β_1. But the two conditional means may need very different specifications, and $\mathbb{E}[Y(1) \mid X] - \mathbb{E}[Y(0) \mid X] = \tau(X)$ is in general a function of X. Concerns: lack of overlap leads to *extrapolation* (if no treated unit has a given X, what does the comparison mean?), and covariate adjustment is usually done while looking at outcomes, which invites cherry-picking.

11.4 Propensity score methods

Motivation. Matching people on many covariates at once is hopeless, since nobody shares all values. The propensity score compresses the covariates into a single number, the probability of treatment, and comparing people with the same score is enough.

The **propensity score** is $e(X_i) = P(Z_i = 1 \mid X_i)$, a one-dimensional summary of possibly many covariates (Rosenbaum and Rubin, 1983). Key result:

Theorem 11.1 (balancing property)

If $(Y(1), Y(0)) \perp\!\!\!\perp Z \mid X$, then $(Y(1), Y(0)) \perp\!\!\!\perp Z \mid e(X)$.

Proof [Supplement]. It suffices to show $P(Z = 1 \mid Y(1), Y(0), e(X)) = e(X)$, which does not depend on the potential outcomes. By the tower property, $P(Z = 1 \mid Y(1), Y(0), e(X)) = \mathbb{E}\big[P(Z = 1 \mid Y(1), Y(0), X) \mid Y(1), Y(0), e(X)\big] = \mathbb{E}\big[e(X) \mid Y(1), Y(0), e(X)\big] = e(X)$, using unconfoundedness in the middle step and that $e(X)$ is a function of X. ■

So we need not find people sharing all covariate values; sharing $e(X)$ is enough. The propensity score is known in a randomised experiment but must be estimated in an observational study, usually by logistic regression, $\operatorname{logit} P(Z_i = 1 \mid X_i) = \beta_X^{\mathsf{T}} X_i$ and $\hat{e}(X_i) = \operatorname{expit}(\hat{\beta}_X^{\mathsf{T}} X_i)$. It is then used for stratification, matching or weighting.

11.4.1 Stratification

Within strata of units with the same $e(X)$, unconfoundedness makes treatment random. Since $\hat{e}$ is continuous, make narrow strata: (1) estimate $\hat{e}(X_i)$; (2) cut it into K strata, e.g. at quantiles or at $(0, \frac{1}{K}], \ldots, (\frac{K-1}{K}, 1)$; (3) treat each stratum as a randomised experiment:

$$\hat{\tau} = \sum_{k=1}^{K} (\bar{Y}_{k1} - \bar{Y}_{k0}) \, \frac{n_k}{n},$$

with $\bar{Y}_{k1}$, $\bar{Y}_{k0}$ the treated and control means in stratum k of size n_k. Equivalently, a treated unit in stratum k has weight $\frac{1}{n_{k1}} \frac{n_k}{n}$ and a control $\frac{1}{n_{k0}} \frac{n_k}{n}$, and a weighted regression $Y = \alpha + \beta Z + \epsilon$ gives $\hat{\tau} = \hat{\beta}$.

11.4.2 Matching

1:1 matching, when there are fewer treated than controls: (1) for each treated unit select a control with the same (or similar) X or $e(X)$, with or without replacement (a bias–variance trade-off); (2) check covariate balance between the treated and matched controls; (3) discard unmatched controls, which often reduces variance; (4) analyse the pairs as a paired randomised experiment (like twins): the mean of treated-minus-control differences estimates the effect (or regress Y on Z in the matched data; adding X is possible but not recommended), and randomisation inference tests the null hypothesis of no effect.

11.4.3 Inverse probability weighting

In a randomised trial the groups are balanced; in an observational study they are not. **IPW** weights treated units by $1/e(X)$ and controls by $1/(1 - e(X))$. Intuition: a treated person with $e(X) = 1/5$ stands for five "clones", one treated and four controls on average, so he represents all five. The weights create a **pseudo-population** in which treatment no longer depends on X: unconfoundedness given X in the original population becomes $(Y(1), Y(0)) \perp\!\!\!\perp Z$ in the pseudo-population.

Proposition 11.2 (IPW identification)

Under unconfoundedness and overlap,

$$\tau = \mathbb{E}\Big[\frac{Z_i Y_i}{e(X_i)}\Big] - \mathbb{E}\Big[\frac{(1 - Z_i) Y_i}{1 - e(X_i)}\Big].$$

Proof. First term: $\mathbb{E}\big[\frac{Z_i Y_i}{e(X_i)}\big] = \mathbb{E}\big[\frac{\mathbb{E}[Z_i Y_i \mid X_i]}{e(X_i)}\big] = \mathbb{E}\big[\frac{e(X_i)\,\mathbb{E}[Y_i \mid Z_i = 1, X_i]}{e(X_i)}\big] = \mathbb{E}[\mathbb{E}[Y_i(1) \mid Z_i = 1, X_i]] = \mathbb{E}[\mathbb{E}[Y_i(1) \mid X_i]] = \mathbb{E}[Y_i(1)]$, using $\mathbb{E}[Z_i Y_i \mid X_i] = P(Z_i = 1 \mid X_i)\mathbb{E}[Y_i \mid Z_i = 1, X_i]$, consistency and unconfoundedness. The second term equals $\mathbb{E}[Y_i(0)]$ in the same way [Supplement]. ■

With estimated scores $\hat{e}_i$,

$$\hat{\tau}^{\mathrm{ATE}} = \frac{1}{n} \sum_i \frac{Z_i Y_i}{\hat{e}_i} - \frac{1}{n} \sum_i \frac{(1 - Z_i) Y_i}{1 - \hat{e}_i},$$

or a regression of Y on Z with weights $w_i = Z_i/\hat{e}_i + (1 - Z_i)/(1 - \hat{e}_i)$. When $\hat{e}_i$ is near 0 or 1 the weights are huge and the estimate unstable. For the effect on the treated (ATT) the weights are 1 for treated units and $\hat{e}_i/(1 - \hat{e}_i)$ for controls.

11.5 Example: job training and earnings

`MatchIt` (matching) and `WeightIt` (weighting) implement these methods; `cobalt::bal.tab()` checks balance and `marginaleffects::avg_comparisons()` estimates effects with appropriate standard errors. The data are `cobalt::lalonde` (185 treated, 429 controls).

```
no.adj <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                  data = lalonde, method = NULL, distance = "glm")
summary(no.adj)$sum.all[, 1:3]
```

```
##            Means Treated Means Control Std. Mean Diff.
## distance    5.774355e-01     0.1822248      1.79408605
## age         2.581622e+01    28.0303030     -0.30944526
## educ        1.034595e+01    10.2354312      0.05496466
## raceblack   8.432432e-01     0.2027972      1.76154189
## racehispan  5.945946e-02     0.1421911     -0.34984254
## racewhite   9.729730e-02     0.6550117     -1.88186760
## married     1.891892e-01     0.5128205     -0.82630927
## nodegree    7.081081e-01     0.5967366      0.24497023
## re74        2.095574e+03  5619.2365064     -0.72108381
## re75        1.532055e+03  2466.4844431     -0.29026291
```

Standardised mean differences before adjustment are huge (1.76 for Black, -1.88 for White, -0.72 for 1974 earnings); a common rule of thumb asks for $|\text{SMD}| < 0.1$ [Supplement]. Three adjustments:

```
ps.str <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                  data = lalonde, method = "subclass", distance = "glm", link = "probit"
    , subclass = 7)
ps.mat <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                  data = lalonde, method = "nearest", distance = "glm", link = "probit",
     replace = TRUE)
W <- weightit(treat ~ age + educ + race + married + nodegree + re74 + re75,
              data = lalonde, method = "glm", estimand = "ATT")
round(cbind(stratified = summary(ps.str, un = FALSE)$sum.across[, 3],
            matched = summary(ps.mat, un = FALSE)$sum.matched[, 3]), 3)
```

```
##            stratified matched
## distance        0.063   0.003
## age            -0.036  -0.085
## educ            0.027  -0.083
## raceblack       0.084   0.015
## racehispan     -0.018   0.046
## racewhite      -0.089  -0.055
## married        -0.030   0.097
## nodegree        0.027   0.143
## re74           -0.089  -0.013
## re75           -0.074  -0.017
```

```
bal.tab(W, un = TRUE)
```

```
## Balance Measures
##                 Type Diff.Un Diff.Adj
## prop.score  Distance  1.7941  -0.0205
## age          Contin. -0.3094   0.1188
## educ         Contin.  0.0550  -0.0284
## race_black    Binary  0.6404  -0.0022
## race_hispan   Binary -0.0827   0.0002
## race_white    Binary -0.5577   0.0021
## married       Binary -0.3236   0.0186
```

```
## nodegree      Binary  0.1114   0.0184
## re74         Contin. -0.7211  -0.0021
## re75         Contin. -0.2903   0.0110
##
## Effective sample sizes
##            Control Treated
## Unadjusted  429.       185
## Adjusted     99.82     185
```

All three bring the covariates close to balance. Stratification uses 7 strata; 1:1 nearest-neighbour matching with replacement uses only 84 distinct controls (the others are discarded); IPW keeps everyone with weights. Effects on the treated, with a weighted regression of 1978 earnings on treatment:

```
str.data <- match.data(ps.str); mat.data <- match.data(ps.mat)
est <- function(fit, ...) {
  a <- avg_comparisons(fit, variables = "treat", ...)
  round(c(estimate = a$estimate, low = a$conf.low, high = a$conf.high))
}
fit.str <- lm(re78 ~ treat, data = str.data, weights = weights)
fit.mat <- lm(re78 ~ treat, data = mat.data, weights = weights)
lalonde$ipw.weight <- W$weights
fit.ipw <- lm(re78 ~ treat, data = lalonde, weights = ipw.weight)
rbind(
  stratification = est(fit.str, vcov = ~subclass, newdata = subset(str.data, treat == 1)
    , wts = "weights"),
  matching       = est(fit.mat, vcov = "HC3", newdata = subset(mat.data, treat == 1),
    wts = "weights"),
  ipw            = est(fit.ipw, vcov = "HC3", wts = "ipw.weight"),
  naive          = round(c(coef(lm(re78 ~ treat, data = lalonde))[2], NA, NA)))
```

```
##               estimate  low high
## stratification     1301 -423 3025
## matching           1290 -603 3184
## ipw                1214 -416 2844
## naive              -635   NA   NA
```

Full matching (each unit in a matched set, no units discarded) gives a larger estimate:

```
ps.full <- matchit(treat ~ age + educ + race + married + nodegree + re74 + re75,
                   data = lalonde, method = "full", distance = "glm", link = "probit")
full.data <- match.data(ps.full)
fit.full <- lm(re78 ~ treat, data = full.data, weights = weights)
est(fit.full, vcov = ~subclass, newdata = subset(full.data, treat == 1), wts = "weights"
    )
```

```
## estimate      low     high
##     1855      364     3346
```

The naive comparison suggests the programme *lowered* earnings; after adjustment the estimates are positive, around \$1 200–1 900. The original randomised experiment found \$2 114 with 95% CI (458, 3955), closest to full matching.

Take-home messages. Matching and weighting separate *design* from *analysis*: neither uses the outcome at the design stage. Matching finds similar subjects with respect to X or $e(X)$; weighting uses inverse probabilities of the treatment received. The analysis is then simple: compare means, or fit a simple regression. Try several methods and check whether they agree.